

Conference Program

November 21-23, 2024

Taipei, Taiwan

Welcome Message

Local Host:

Dr. Li, Kun Chung

Dr. Li has an extensive and distinguished career in the field of education. He currently serves as the Chair Professor and Vice President at the Department of Early Childhood Education, Nanhua University, Taiwan. Previously, he held the position of Executive Secretary at the Ministry of Education and was the Director of the Preparatory Office at the National Academy of Educational Research. Additionally, he served as Professor and Director at the Graduate Institute of Education at National Cheng Kung University. At the Southern Taiwan University of Science and Technology, he was both the Vice President and a Professor at the Graduate Institute of Educational Leadership and Evaluation. He also worked as the Executive Vice President at Quanzhou Ocean Vocational College.



Dear Participants and Guests,

Welcome to Taiwan, a beautiful island full of charm and diverse cultures. We are honored to host this conference here and eagerly look forward to discussing various important topics with you.

Taiwan boasts stunning natural landscapes, from the spectacular Taroko Gorge and the mysterious Alishan Forest to the beautiful Sun Moon Lake, all of which will captivate you. Additionally, the magnificent views from Taipei 101 are not to be missed, where you can overlook the entire city of Taipei.

Besides its natural beauty, Taiwan is also renowned for its world-famous cuisine. The night market culture is a unique feature of Taiwan, where you can taste a variety of snacks such as oyster omelets, bubble tea, beef noodles, and stinky tofu. Each dish is sure to delight your taste buds. Taiwan's food is not only diverse but also uniquely flavored, making it a must-try experience.

We hope that during this conference, you will have the opportunity to explore Taiwan's culture and scenery and enjoy a wonderful time here. May you not only gain insights from the conference but also relish the unique charm of this beautiful island.

Once again, welcome to Taiwan, and may every moment of your stay be filled with joy and surprises!

Sincerely,

Li, Kun Chung

International Committees

A. W Jayawardena , The University of Hong Kong
Ahmad Mahdavi , University of Tehran
Amy Lee , Hong Kong Metropolitan University
Angela Gandra , Mackenzie Presbyterian University
Ann Mary , Teresian College
April Aura A. Cacho , Northern Iloilo State University
Bob Barrett , WEB School of Business
Cheng-I Chen , National Central University
Chih-Ying Lee , Nanhua University
Chin-Yi Fang , National Taiwan Normal University
Chung Hwei Su , National Kaohsiung University of Science
and Technology
Elcid Bautista , PAREF Southridge School
Hitoshi Sasaki , Takushoku University
Hye-Sook Park , Honam University
Johan Van Rooyen , Webster University
Jonald De Guzman , Bigo Elementary School
Jong-Dar Yau , Tamkang University
José Vargas Hernandez , University of Guadalajara
Jung Yun Hur , California State University
K M Metry , Kannada University
Kuang-Cheng Chen , National Taipei University of Technology
Kun-Chung LI , Nanhua University

Lauren Suzanne Power , Sophia University
Madhusudan Datta , University of Kalyani
Marilyn Obod , World Citi Colleges
Michelle Kawamura , Ritsumeikan University
Mohamad Hanapi , Asia e University
Nam Chol Yu , KimChaek University of Technology
Neal Luo , Emporia State University
Pei-Chun Lin , National Cheng Kung University
Pranati Panda , National Institute of Educational Planning and
Administration
Sai Sathish Kethu , NeuraFlash
Sangho Kim , Ritsumeikan Asia Pacific University
Vida D. Jimenez , Philippine Christian University
Vijayakumar Babuavadhanula , Osmania University
Warner Woodworth , University of Utah
Wen-Tsai Sung , National Chin-Yi University of Technology
Zonghie Han , Daegu University

General Information for Participants

■ Registration

The registration desk will be situated on the **B1F** at the **Sheraton Grand Taipei Hotel** during the following time:

08:30-15:00 Thursday, November 21, 2024

08:30-12:00 Friday, November 22, 2024

■ A Polite Request to All Participants

Participants are requested to arrive in a timely fashion for all addresses, whether to their own, or to those of other presenters. Presenters are reminded that the time slots should be divided fairly and equally between the number of presentations, and that they should not overrun. The session chair is asked to assume this timekeeping role and to summarize key issues in each topic.

■ Preparation for Oral Presentations

All presentation rooms are equipped with a screen, an LCD projector, and a laptop computer installed with Microsoft PowerPoint. You will be able to insert your USB flash drive into the computer and double check your file in PowerPoint. We

recommend you to bring two copies of the file in case that one fails. You may also connect your own laptop to the provided projector; however please ensure you have the requisite connector.

Preparation for Poster Presentation

Materials Provided by the Conference

Organizer:

1. X-frame display & base fabric canvases (60cm×160cm)
2. Adhesive tapes or binder clips



Materials Prepared by the Presenters:

1. Home-made Poster(s)
2. Material: not limited, can be posted on the canvases
3. Recommended poster size: 60cm*160cm

Special Thanks to Session Chairs

Dr. Lee Chih Ying, *Nanhua University*

Prof. Li, Kun Chung, *Nanhua University*

Prof. Farn Wang, *National Taiwan University*

A.P. Alice Yin Wa Chan, *City University of Hong Kong*

Prof. Hiroshi Miyazaki, *Kansai University*

Prof. Rafal Obuchowicz, *Jagiellonian University*

Prof. Kuang-Cheng Chen, *National Taipei University of Technology*

A.P. Krawee Treeamnuk, *Suranaree University of Technology*

Prof. Michiko Miyamoto, *Nagasaki University*

A.P. Tawarat Treeamnuk, *Suranaree University of Technology*

Conference Venue Information

Sheraton Grand Taipei Hotel

No 12, Sec 1, Zhongxiao East Road, Zhongzheng District,

Taipei City, Taiwan, 10049

Phone: +886 2-23215511

Location

The hotel boasts a convenient location for a quick access to transportation and shopping districts. Restaurant was awarded 2 Michelin star and Michelin recommendations.

Transportation

A. Taipei Songshan Airport MRT

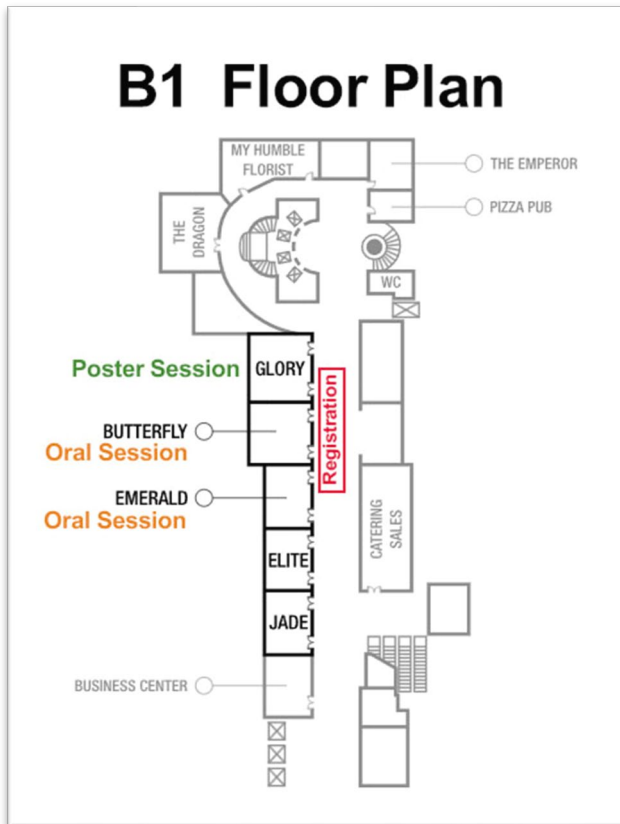
Take the Brown Wenhua Line at Songshan Airport Station to ZhongxiaoFuxing Station, transfer to the Blue Bannan Line to Shandao Temple Station (Exit 2 Shandao Temple MRT Station)

B. Taiwan Taoyuan International Airport MRT

From Taoyuan Airport, you can take the Airport MRT to Taipei Main Station. From there, you can either take the MRT Blue

Line to Shandao Temple Station or walk for 8 minutes from Exit M8 of Taipei Main Station to reach the Sheraton Grand Taipei Hotel.

B1 Floor Plan



Oral Sessions: Butterfly, Emerald

Poster Session: Glory

Lunch: Kitchen 12 (1 Floor)

Conference Schedule

Thursday, November 21, 2024

Oral Sessions

Sheraton Grand Taipei Hotel , B1F

Time	Schedule	Venue
08:30-15:00	Registration	Foyer area
09:15-10:45	Education, Psychology & Society	Emerald
	Environmental Engineering	Butterfly
10:45-11:00	Tea Break & Networking	Foyer Area
	● Local Host Welcome Speech ● Keynote Address: <u>Prof. Li, Kun Chung</u>	
11:00-12:00	Topic: Five Steps of Teaching with Generative AI ● Keynote Address: <u>Dr. Chih Ying Lee</u>	Emerald

Topic: Designing and Quality Assurance of Distance Learning Courses: A Case Study of the E-Learning In-Service Master Program of Humanistic Buddhism at the College of Humanities

- **Keynote Address:**
Prof. Farn Wang
Topic: As AI is Competing for Programmers' Jobs, What Shall We Programmers Do? From A Software Testing Researcher's Peerspective

Butterfly

12:00-13:30	Lunch Time	Kitchen 12
13:30-15:00	Information Sciences & Special Session-Interdisciplinary Innovations: Exploring Semiconductors, AI, and Sustainability	Butterfly
15:00-15:20	Tea Break & Networking	Foyer Area

Thursday, November 21, 2024

Poster Sessions

Sheraton Grand Taipei Hotel , B1F

Time	Schedule	Venue
08:30-15:00	Registration	Foyer area
10:00-11:00	Education, Information Sciences, Civil Engineering, Mechanical Engineering, Civil Engineering, Mechanical Engineering, Computer Engineering and Technology, Material Science and Engineering	Glory
10:45-11:00	Tea Break & Networking	Foyer Area
12:00-13:30	Lunch Time	Kitchen 12

Friday, November 22, 2024

Oral Sessions

Sheraton Grand Taipei Hotel , B1F

Time	Schedule	Venue
08:30-12:00	Registration	Foyer area
09:00-10:30	Law, Politics and Public Administration	Emerald
	Industrial Engineering, Fundamental and Applied Sciences, Mechanical Engineering	Butterfly
10:30-10:45	Tea Break & Networking	Foyer Area
10:45-12:15	Business & Management	Emerald
	Biological Engineering	Butterfly
12:00-13:30	Lunch Time	Kitchen 12

Saturday, November 23, 2024

Committee Meeting (Internal Only)

Keynote Address

Emerald

November 21, Thursday 11:05-11:30

Topic: Designing and Quality Assurance of Distance Learning Courses: A Case Study of the E-Learning In-Service Master Program of Humanistic Buddhism at the College of Humanities

Dr. Lee Chih Ying

Graduate Institute of Religious Studies

Nanhua University

Abstract:

To embrace the era of digital learning, the Ministry of Education has meticulously planned and proposed the comprehensive "Digital Learning Promotion Plan." Additionally, the Executive Yuan has put forth a long-term plan titled "Digital Nation and Innovative Economic Development Program," aiming to realize the vision of a "Digital Nation, Smart Island" In Taiwan.

Nanhua University has established the "E-Learning In-Service Master Program of Humanistic Buddhism at the College of

Humanities" to actively align with national policies. This program is specially designed with the concept of continuing education and digital learning methods. Through distance learning courses, it helps global learners in continuing education to overcome spatial and temporal barriers, enhancing their professional abilities and knowledge in Humanistic Buddhism. It broadens their perspectives and views on the study of Humanistic Buddhism, fostering researchers who emphasize both theory and practice and have a comprehensive understanding of Humanistic Buddhism. This program aims to cultivate professionals who can effectively utilize digital technology and integrate it with their specialized skills in the research of Humanistic Buddhism.

Furthermore, a quality assurance system for distance learning courses has been developed to establish the effectiveness of teaching and learning in distance education. This report will explain how distance learning courses are planned, designed, implemented, and quality controlled, using the digital learning courses in Humanistic Buddhism at Nanhua University as an example, providing a reference for distance learning.

Keynote Address

Emerald

November 21, Thursday 11:30-12:00

Topic: Five Steps of Teaching with Generative AI

Prof. Li, Kun Chung

Department of Early Childhood Education,
Nanhua University

Abstract

This report aims to elucidate the five steps of teaching with generative AI, assisting teachers in mastering the teaching principle of "use wisely, not prohibit; an assistant, not a ghostwriter." The five steps of teaching with generative AI are "Preparation, Prompt, Verification, Integration, and Optimization."

1. Preparation involves comprehensive readiness, adhering to the premise that "good preparation leads to good questioning, and asking the right questions is more important than providing the right answers." This step encourages students to awaken old

experiences or prior knowledge, collect and read new information, and form preliminary ideas.

2. Prompt focuses on precise inquiry. Questions should fit the role scenario, task theme, action level, and output requirements, and provide appropriate guiding examples or advanced inquiries. Teachers must guide students in developing a questioning scaffold.

3. Verification entails critical examination, specifically scrutinizing the information provided by generative AI for accuracy or errors, consistency of content, relevance to related information, comprehensiveness, bias, and potential violations of privacy and intellectual property rights.

4. Integration represents diverse consolidation, integrating the results of multiple uses of generative AI with the knowledge (or skills and attitudes), academic journal articles, and peer discussions taught by teachers. It can consider both horizontal and vertical integration, adopting methods from physical to chemical integration.

5. Optimization/Annotation involves optimizing the integrated results and appropriately annotating and citing them to avoid

plagiarism and infringement, striving for innovation and improvement. It emphasizes strengthening the PDCA (Plan-Do-Check-Act) quality assurance cycle and competency-oriented teaching.

These steps guide teachers in effectively leveraging generative AI in education, emphasizing preparation, critical thinking, and integration to enhance teaching and learning outcomes.

Keynote Address

Butterfly

November 21, Thursday 11:10-11:30

***Topic: As AI is Competing for Programmers' Jobs,
What Shall We Programmers Do? From A Software
Testing Researcher's Peerspective***

Prof. Farn Wang

Dept. of Electrical Engineering

National Taiwan University

Abstract

KOL are warning that programmers' job opportunities will be eliminated by AI. More and more development teams prompt AI for program templates.

In this trend, although less and less programmers will be needed, job opportunities for requirement engineers and program QA engineers will become stronger than ever.

In this speech, we discuss how software test teams may adapt to and benefit from the huge opportunity of AI technology.

Finally, we present a new product concept and a business model for software testing service in the AI era.

Thursday, November 21

Oral/ Poster Sessions

Oral Sessions

Education, Psychology & Society

Thursday, November 21 09:15-10:45 Emerald

Session Chair: A. P. Alice Yin Wa CHAN

ISESS-0157

What Are Students' Perceptions of the Advantages and Challenges of a Digital Research-based Collaborative Project?

Alice Yin Wa CHAN | *City University of Hong Kong*

Chit Cheung Matthew SUNG | *City University of Hong Kong*

ISESS-0174

Daily Music Listening to Enhance Resilience among Secondary School Students: A Positive Psychology Perspective

Lai Yin Sarah Wan | *The Education University of Hong Kong*

Yuen Man Ho | *The Education University of Hong Kong*

ISESS-0160

Exploring the Mental Health Training Needs of Pre-Service Teachers in Hong Kong

Arita Wing Yan Chan | *The Education University of Hong Kong*

ICSSAM-0058

Exploring Relevance of Walkability in a Digitalized and High-Density City: Experiences of Older Adults in Singapore

Eng Fei, Koh | *Singapore University of Technology and Design*

ISESS-0159 (Online Presentation)

Development and Validation of Self-Reliance Behaviors Among Elderly in Multicultural Societies: A Study of Measurement Invariance

Kriangsak Rattakul | *Songkhla Rajabhat University*

Thawatchai Sripornngam | *Songkhla Rajabhat University*

Warakorn Supwirapakorn | *Burapha University*

Environmental Engineering

Thursday, November 21 09:15-10:45 Butterfly

Session Chair: Prof. Hiroshi MIYAZAKI

GCEAS-11778

Investigation of Heat Stress Mediated Heat Stroke Risk Area Extraction Methods

Soshi FUJIHARA | *Kansai University*

Hiroshi MIYAZAKI | *Kansai University*

GCEAS-11780

Risks of Evacuation Behavior in Hot Summer Daytime

Naho ITANI | *Kansai University*

Soshi FUJIHARA | *Kansai University*

Hiroshi MIYAZAKI | *Kansai University*

GCEAS-11779

Proposal of a Correction Method for Surface Temperature Using the Water Vapor Image of the Himawari AHI Sensor

Jin SHIMIZU | *Kansai University*

Hiroshi MIYAZAKI | *Kansai University*

GCEAS-11776

**Research on Prediction of Climate Change Impacts
Considering Hourly Temperature -Development of the
Hourly Temperature Estimation Model Using Machine
Learning-**

Ryuhei MATSUKI | *Kansai University*

Hiroshi MIYAZAKI | *Kansai University*

**Information Sciences & Special Session-
Interdisciplinary Innovations: Exploring
Semiconductors, AI, and Sustainability**

Thursday, November 21 13:30-15:00 Butterfly

Session Chair: Prof. Rafal Obuchowicz

ICEAI-0177

**Distributed AI with Autoencoders for Large Medical Dataset
Processing: The Master Collector Concept**

Rafal Obuchowicz | *Jagiellonian University*

Piotr Reichert | *Raygenic Technology*

Robert Ratajczak | *Raygenic Technology*

Grzegorz Materna | *Raygenic Technology*

Michal Trojanowski | *Raygenic Technology*

Michal Strzelecki | *Lodz University of Technology*

ICEAI-0187

**Comparative Analysis of RNN, LSTM, and GRU Models for
Stock Price Prediction: A Study on Indonesian Blue-Chip
Stocks**

Luciana Abednego | *Parahyangan Catholic University*

Cecilia E. Nugraheni | *Parahyangan Catholic University*

Ni Luh Melika | *Parahyangan Catholic University*

ICEAI-0186

Two-stage No-wait Flexible Flow Shop Scheduling with Heterogeneous Machines Using Firefly Algorithm

Cecilia E. Nugraheni | *Parahyangan Catholic University*

L. Abednego | *Parahyangan Catholic University*

M. Widyarini | *Parahyangan Catholic University*

ICEAI-0178

Smart Agriculture Using IOT and Its Enabled Technologies

Senthilrajan Agniraj | *Alagappa University*

ICEAI-0175

Using Smaller Lookup Table in SubBytes Transformation for IoT Systems

Yan-Haw Chen | *I-Shou University*

Shui-Hsiang Su | *I-Shou University*

Fu-Jung Kan | *I-Shou University*

Hua-You Lin | *I-Shou University*

Hao-Ni Wang | *I-Shou University*

Poster Sessions

**Education, Information Sciences, Civil Engineering,
Mechanical Engineering, Civil Engineering,
Mechanical Engineering, Computer Engineering
and Technology, Material Science and Engineering**
Thursday, November 21 10:00-11:00 Glory

ICSSAM-0054

**A Strategy of Designing Integrated Lessons for Pre-service
Teachers**

SeonMee Kim | *Sunchon National University*

ISESS-0176

**A Systematic Review of Eye Tracking Studies on the Impact
of Instructor Presence in Educational Videos**

Yen Ying Ng | *Nicolaus Copernicus University in Torun*

Noah L. Schroeder | *University of Florida*

ICEAI-0188

**Development of a Missing Person Tracking Solution through
the Design of a Selective Person Re-Identification System**

Chorong Kim | *Innodep Inc.*

ICEAI-0180

Development of a Psychoacoustic Assessment Model for Air-conditioning Building Environments

Kuen Wai Ma | *The Hong Kong Polytechnic University*

Cheuk Ming Mak | *The Hong Kong Polytechnic University*

Fu Lai Korris Chung | *The Hong Kong Polytechnic University*

Hai Ming Wong | *The University of Hong Kong*

ICEAI-0179

Fundamental Research on the Improvement of Operational Efficiency of Reverse Osmosis (RO) Systems Using Nanobubbles

Chia-Lung Kuo | *National Yunlin University of Science and Technology*

Yuan-Jen Chang | *National Taipei University of Technology*

GCEAS-11786

Development of Damage Evaluation System by Cumulative External Forces Using Mechanoluminescence

Mutsumi MIYAGAWA | *Maebashi Institute of Technology*

Jyo SHIMURA | *Takushoku University*

Toru SASAKI | *National Institute of Technology, Nagaoka College*

GCEAS-11785

Optimal Design Method for GFRP/Metal Dissimilar Adhesive Joints Using Additive Manufacturing Technology

Toru SASAKI | *National Institute of Technology, Nagaoka
College*

Takeshi TANE | *National Institute of Technology, Kitakyushu
College*

Mutsumi MIYAGAWA | *Maebashi Institute of Technology*

GCEAS-11783

Deep Learning based Identification of Severe Sleep Apnea using Sagittal MRI Head images

Tsuyoshi Mikami | *National Institute of Technology
Tomakomai College*

Kazuya Yonezawa | *Hakodate National Hospital*

GCEAS-11775

Efficient Oil-Water Separation System with One-step Electrodeposited Superhydrophobic-Oleophilic and Oleophobic-Hydrophilic Metal Mesh Surfaces

Xie Xinying | *Hong Kong Baptist University*

Hu Chuanbo | *Chongqing Three Gorges University*

Ren Kangning | *Hong Kong Baptist University*

Friday, November 22

Oral/ Poster Sessions

Oral Sessions

Law and Public Administration & Politics

Friday, November 22 09:00-10:30 Emerald

Session Chair: Prof. Kuang-Cheng Chen

ICSSAM-0046

Parallel Importation Occurred in Digital Market—From the Perspectives of International Exhaustion Doctrine and Territoriality Principle

Kuang-Cheng Chen | *National Taipei University of Technology*

ISESS-0173

Empowering Community to Alleviate Poverty: A Case Study on Food Security in Don Mod Daeng District, Ubon Ratchathani province

Chantha Thanawattanawong | *Ubon Ratchathani University*

Piyamart Tupmongkol | *Ubon Ratchathani University*

ISESS-0169

From Loudspeakers to Smartphones: The Role and Challenges of Community Leaders in Implementing the ThaID Application in Thailand

Warut Ingkhathawornwong | *Ubon Ratchathani University*

Siriporn Jantanasakulwong | *Ubon Ratchathani University*

ISESS-0165

Enhancing Local Fiscal Efficiency: The Development of Tax Maps and Property Registers by Local Administrative Organizations in Thailand

Darunee Pumkaew | *Ubon Ratchathani University*

ISESS-0178

The Moral Trial: Power Dynamics in Chinese Platform Governance

Yinyu Wu | *Hong Kong University of Science and Technology*

Sirui Han | *Hong Kong University of Science and Technology*

**Industrial Engineering, Fundamental and Applied
Sciences & Mechanical Engineering**
Friday, November 22, 09:00-10:30 Butterfly
Session Chair: A.P. Krawee Treeamnuk

GCEAS-11788

**Investigating the Application of Induction Heating for Hot
Air Production**

Wasawat Thingborirak | *Suranaree University of Technology*

Krawee Treeamnuk | *Suranaree University of Technology*

Tawarat Treeamnuk | *Suranaree University of Technology*

GCEAS-11790

**Prediction Performance of Deep Learning Models for
Estimating Hydrodynamic Characteristics of Fishing Vessels**

Janghoon Seo | *Tongmyong University*

Dong-Woo Park | *Tongmyong University*

Kidong Lee | *Korea Maritime Transportation Safety Authority*

ICEAI-0184

**Real-time Scheduling of Large-scale Manufacturing Systems
Using Reinforcement Learning**

Chae-Won Lee | *Myongji University*

In-Beom Park | *Myongji University*

Business & Management

Friday, November 22 10:45-12:15 Emerald

Session Chair: Prof. Michiko Miyamoto

ICSSAM-0061

**AI Adoption, Strategic Advantage, and IT Governance
Amidst the Pandemic**

Michiko Miyamoto | *Nagasaki University*

ICSSAM-0055

**The Impact of Customer Experience on Consumer
Impulsive Buying with AI-Powered Chatbot
Communication - The Moderating Role of Promotion Focus**

Min-Shi Liu | *National Chung Cheng University*

Le-Chi Chen | *Taipei Fuhsing Private School*

ICSSAM-0050

**The Impact of Variety-Seeking, Status Consumption on
Consumer Purchase Intention in Cross-Border E-Commerce
- The Moderating Role of Promotion Focus**

Min-Shi Liu | *National Chung Cheng University*

Fannie Cheng | *Taipei American School*

ICSSAM-0047

Justice Within and Between Teams

Nathan Tong | *ESSCA School of Management*

Biological Engineering

Friday, November 22 10:45-12:15 Butterfly

Session Chair: A. P. Tawarat Treeamnuk

GCEAS-11789

Inactivation of Microbial Cells in Green Oak Lettuce Using Plasma Activated Water

Kamontip Preeprem | *Suranaree University of Technology*

Tawarat Treeamnuk | *Suranaree University of Technology*

Krawee Treeamnuk | *Suranaree University of Technology*

Wichannarong Wongbubpa | *Suranaree University of Technology*

Kajondet Labut | *Suranaree University of Technology*

GCEAS-11787

Paddy Dehumidification with Obstructed Grain Flow in Infrared Dryer

Wichannarong Wongbubpa | *Suranaree University of Technology*

Krawee Treeamnuk | *Suranaree University of Technology*

Tawarat Treeamnuk | *Suranaree University of Technology*

Kamonthip Preeprem | *Suranaree University of Technology*

Nuttapong Wongbubpa | *Rajamangala University of Technology Isan*

GCEAS-11782

**The Optimization of Vacuum Frying Conditions for Nam
Dok Mai Mango using Response Surface Method**

Woraphon Denduang | *Suranaree University of Technology*

Tawarat Treeamnuk | *Suranaree University of Technology*

Krawee Treeamnuk | *Suranaree University of Technology*